

Theoretical Foundations for 3D Reconstruction of Sports Games with Limited Visual Data Set- tings

Independent Study Course (NST3901)

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Motivation

Semi-automated offside technology



VAR controversies in the 2024 ASEAN Cup



A suggested pipeline

3-stage pipeline

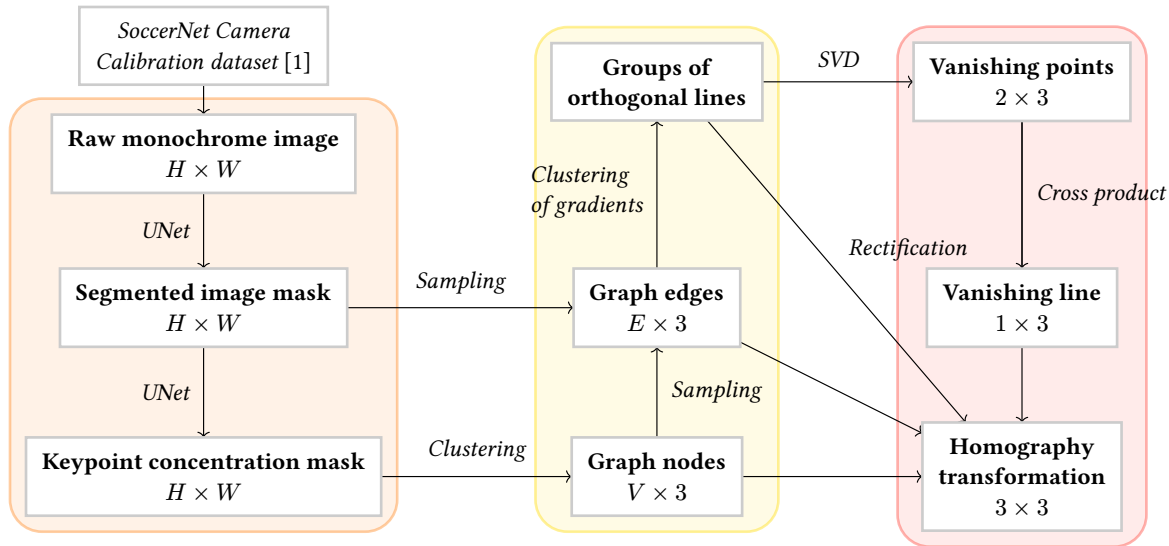
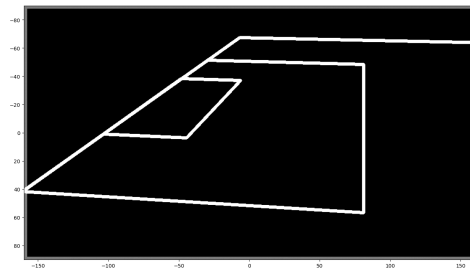
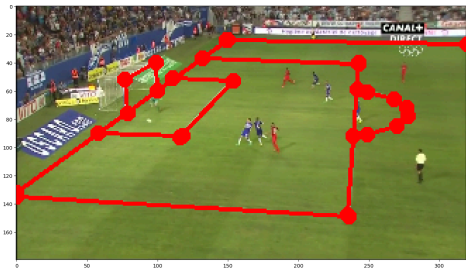
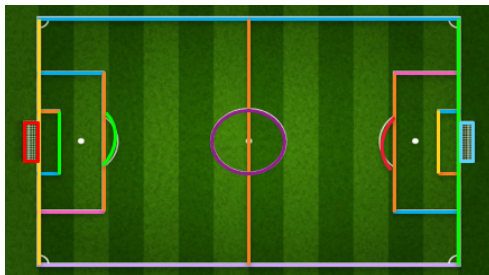


Figure 3: Summary of the suggested automated pipeline for field registration

3-stage pipeline

- **Feature masks generation.**
 - Generative image models (U-Net [2])
 - Gray-scale image masks
- **Graph creation.**
 - Keypoints, field lines $\implies$ Graph
 - **Orthogonal** field lines
 - Robustness (gamma correction, sampling)
- **Homography reconstruction.**
 - Projective / Multiview geometry [3]
 - Linear algebra

Dataset



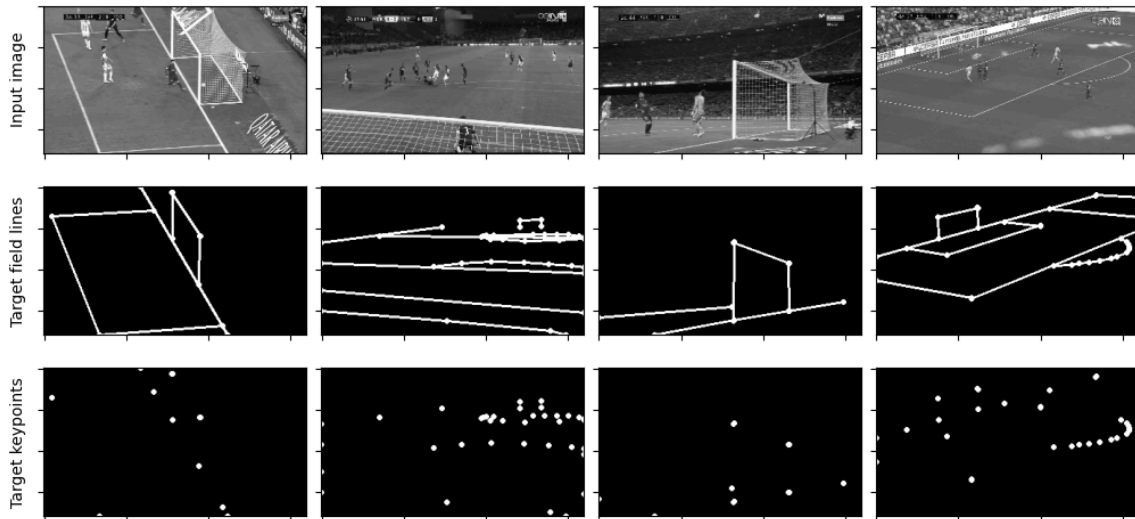
Dataset (ii)

Task / Dataset name	Input	Output / Labels
<i>General</i> CalibrationDataset	Raw image $(3 \times H \times W)$	Raw JSON dictionary
<i>Edge segmentation</i> MaskingCalibrationDataset	Grayscaled image $(H \times W)$	Image mask, highlighted field lines $(H \times W)$
<i>Keypoint concentration</i> MaskingCalibrationDataset (keypoint_only=True)	Grayscaled image $(H \times W)$	Image mask, highlighted keypoints $(H \times W)$
<i>Keypoints</i> PitchKeypointCalibrationDataset	Grayscaled image $(H \times W)$	List of all keypoints, reconstructed via homography (27×3)
<i>Homography reconstruction</i> PitchHomographyCalibrationDataset	Grayscaled image $(H \times W)$	The homography matrix (3×3)

Stage 1. Image segmentation

Segmentation of field lines and keypoints

Segmentation of field lines



UNet [2]: A generative model for feature extraction

- Strong in **localisation** tasks
 - Convolutional layers
 - Detecting patterns based on the relationship between neighbouring pixels
- Skipping neurons
 - Perserves global properties
- Won ISBI 2015 EM segmentation challenge in low-data conditions
 - Ability to learn even with the lack of data

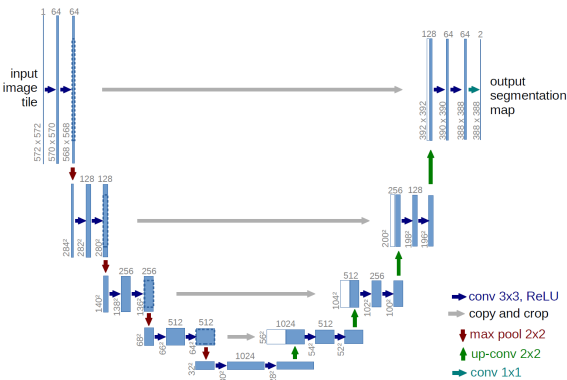


Figure 6: UNet's architecture [2]

Choosing a loss function for segmentation tasks [4]

- Cross Binary Entropy Loss

$$\mathcal{L}_{\text{BCE}}(\vec{x}, \vec{y}) = - \sum_i \log(x_i y_i) = -\log(\vec{x}) \cdot \log(\vec{y})$$

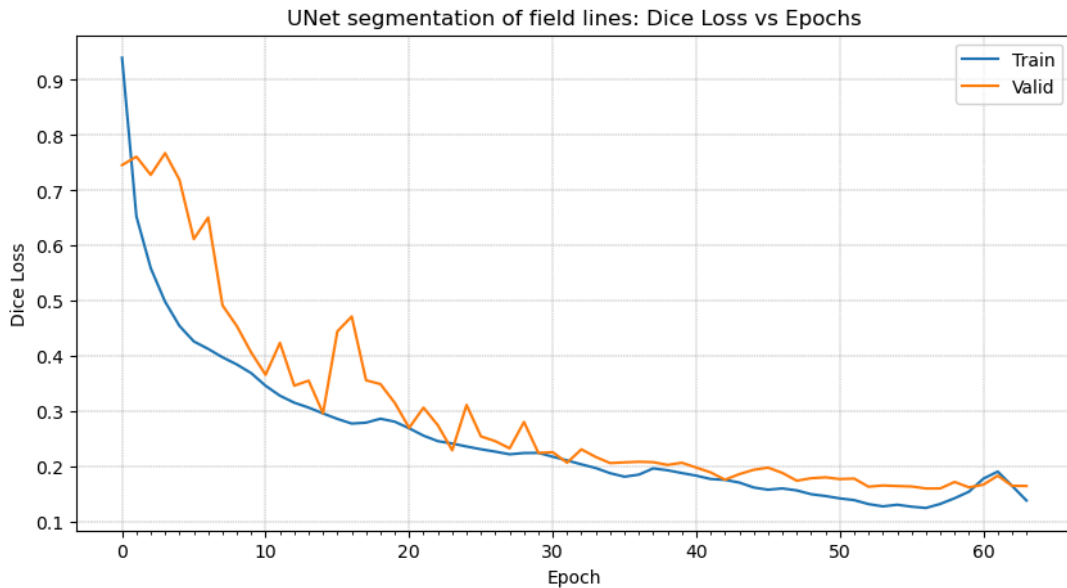
- Intersection-over-Union (IOU) Loss

$$\mathcal{L}_{\text{IOU}}(\vec{x}, \vec{y}) = \frac{|X \cap Y|}{|X \cup Y|} = \frac{\vec{x} \cdot \vec{y}}{1 - (\vec{1} - \vec{x})(\vec{1} - \vec{y})}$$

- Dice Loss (for binary classes)

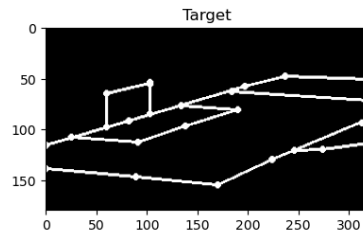
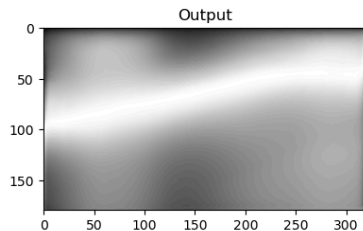
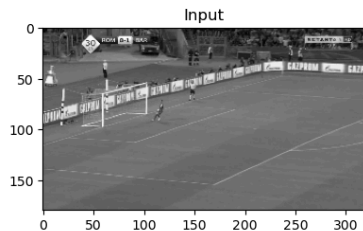
$$\mathcal{L}_{\text{Dice}}(\vec{x}, \vec{y}) = \frac{2|X \cap Y|}{|X| + |Y|} = 2 \cdot \frac{\vec{x} \cdot \vec{y}}{\vec{x} \cdot \vec{1} + \vec{y} \cdot \vec{1}}$$

Training metrics (field lines, $n_epochs = 64$)



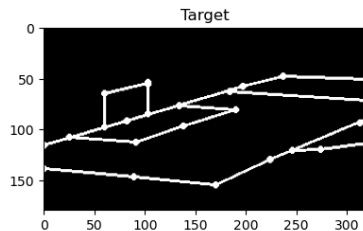
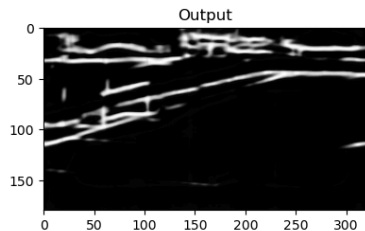
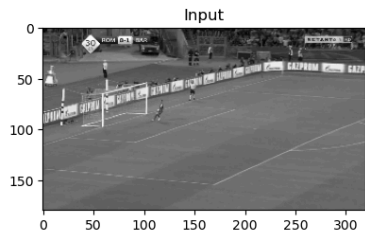
Qualitative results (field lines, epoch = 05)

UNet Segmentation Model. Epoch 5. Test loss: 0.7192678451538086



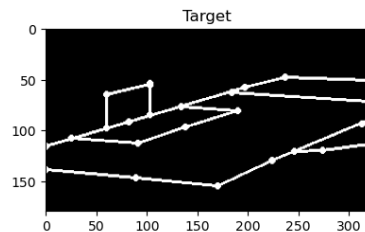
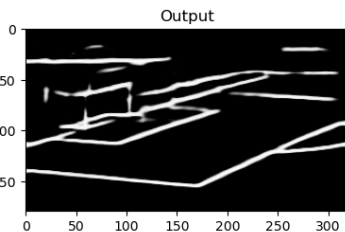
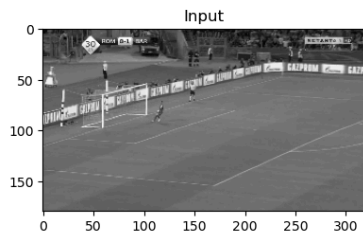
Qualitative results (field lines, epoch = 13)

UNet Segmentation Model. Epoch 13. Test loss: 0.3468126654624939



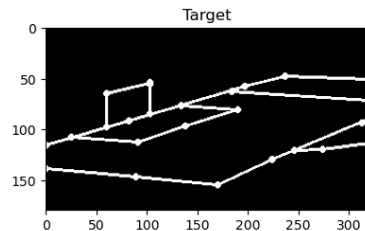
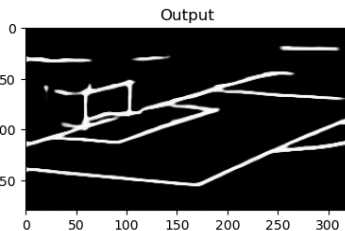
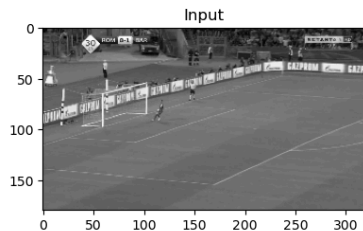
Qualitative results (field lines, epoch = 28)

UNet Segmentation Model. Epoch 28. Test loss: 0.23337285965681076



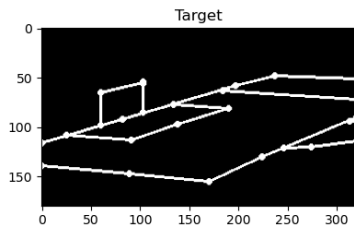
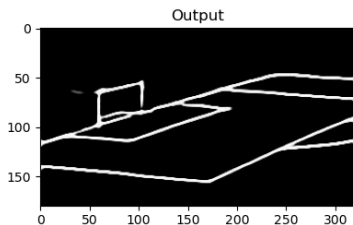
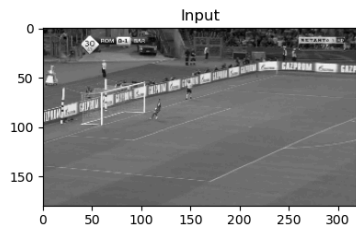
Qualitative results (field lines, epoch = 46)

UNet Segmentation Model. Epoch 46. Test loss: 0.19854416698217392

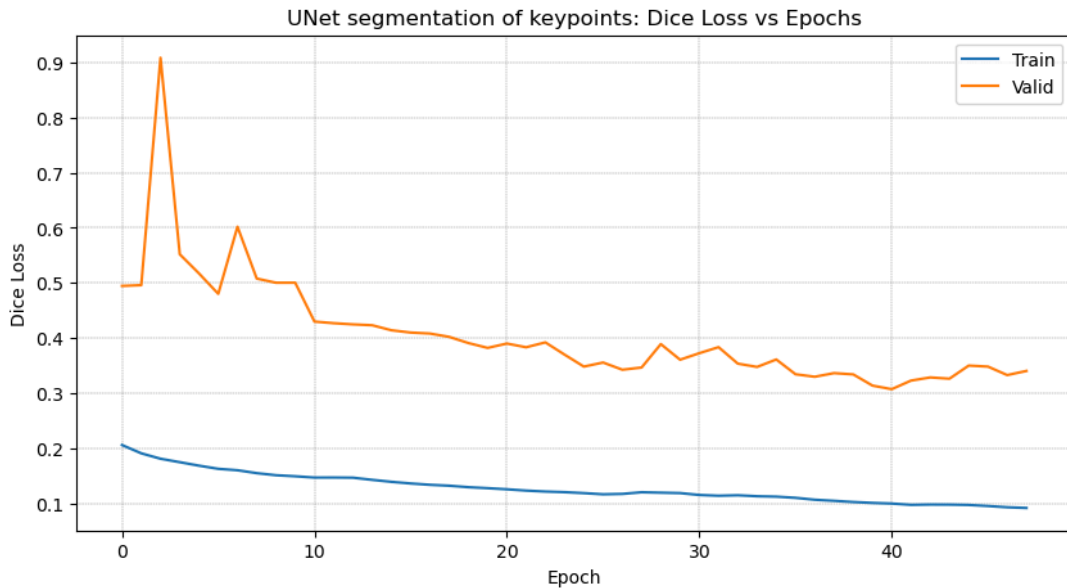


Qualitative results (field lines, epoch = 64)

UNet Segmentation Model. Epoch 64. Test loss: 0.16530044376850128

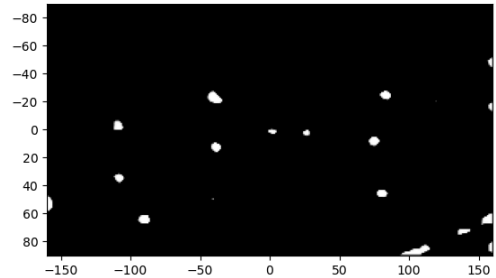
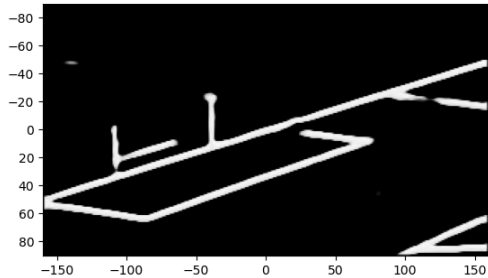


Training metrics (keypoints, n_epochs = 48)

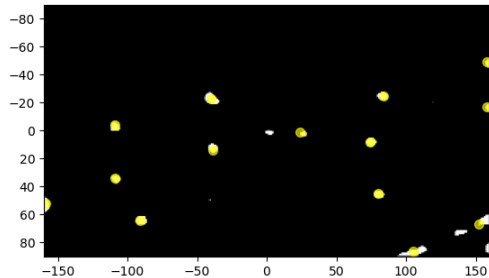
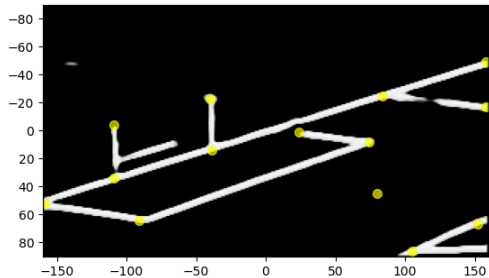


Stage 2. Graph creation

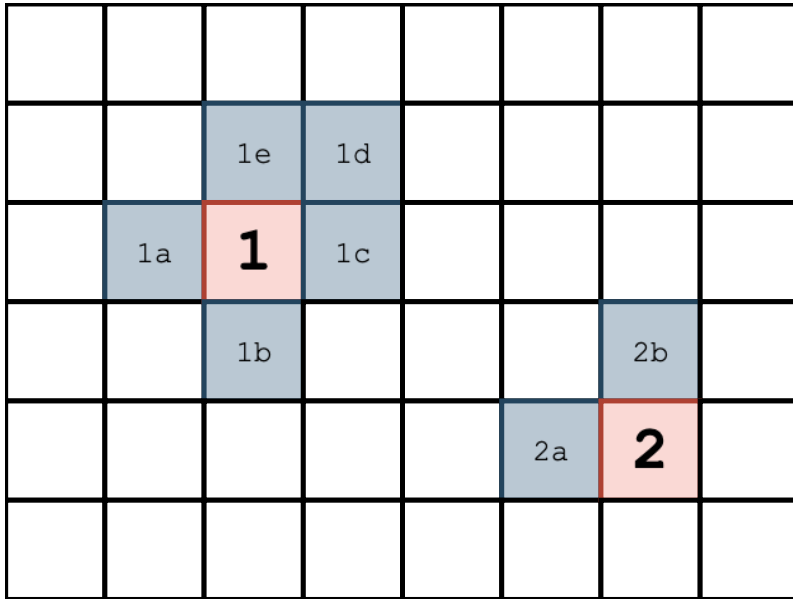
Keypoint detection via pixel clustering



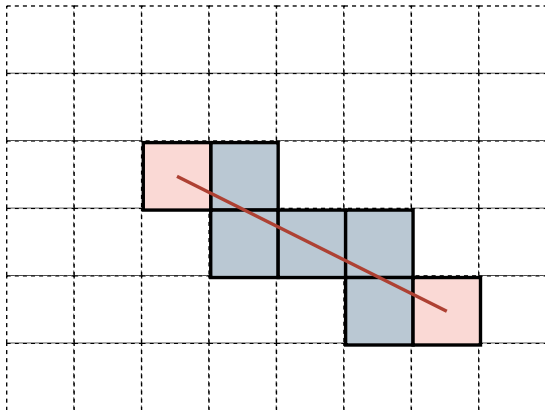
Keypoint detection via pixel clustering



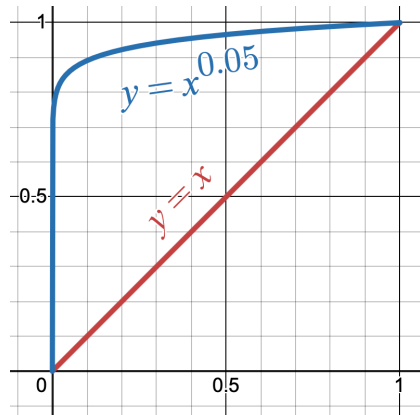
Keypoint detection via pixel clustering



Edge sampling



- Edge mask I_E of size $H \times W$;
- Two keypoints $\vec{p}_1$ and $\vec{p}_2$;



Gamma correction plot, $\gamma = 0.05$

Edge sampling (ii)

- $\gamma \approx 0.01$

Step 1. Let $T := \lceil \|\vec{p}_1 - \vec{p}_2\| \rceil$.

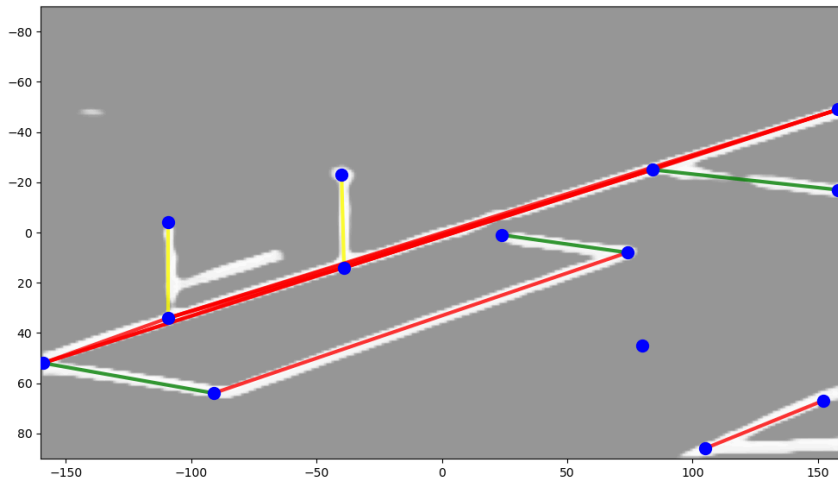
Step 2. Generate T random values of $t^{(i)} \in [0, 1]$.

Step 3. For each $t^{(i)}$, the sampled point is $\vec{q}^{(i)} := (1 - t^{(i)})\vec{p}_1 + t^{(i)}\vec{p}_2 + \vec{\epsilon}$, where $\vec{\epsilon}$ is a random noise vector, taking range $[-2, 2]$ for each dimension.

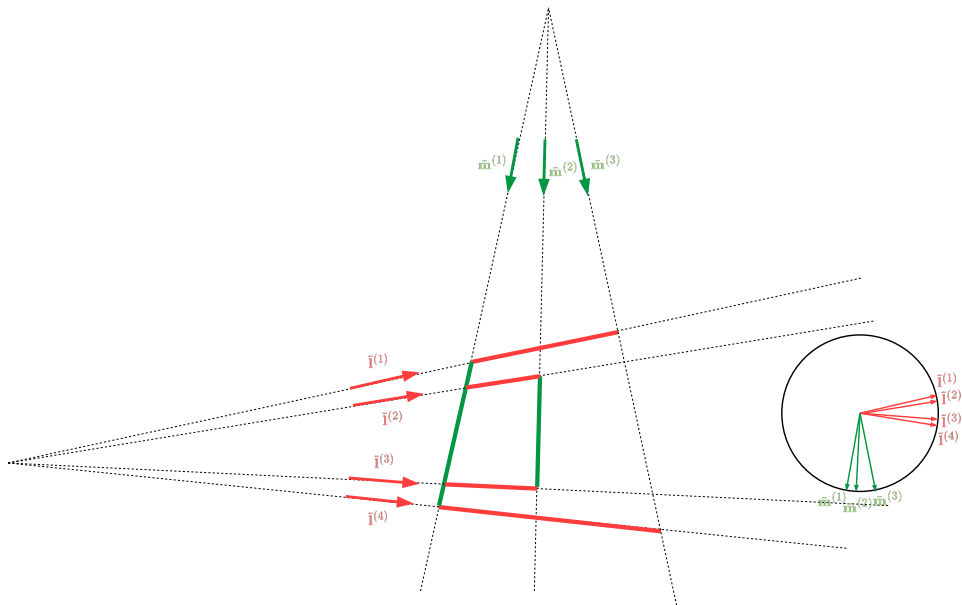
Step 4. Return confidence as the sum of the points' intensities raised to the power of γ :

$$\text{Confidence} := \frac{1}{T} \sum_i I_E(q_x^{(i)}, q_y^{(i)})^\gamma$$

Clustering of gradients



Clustering of gradients (ii)



Clustering of gradients (iii)

Grouping lines by the gradient vector.

Input.

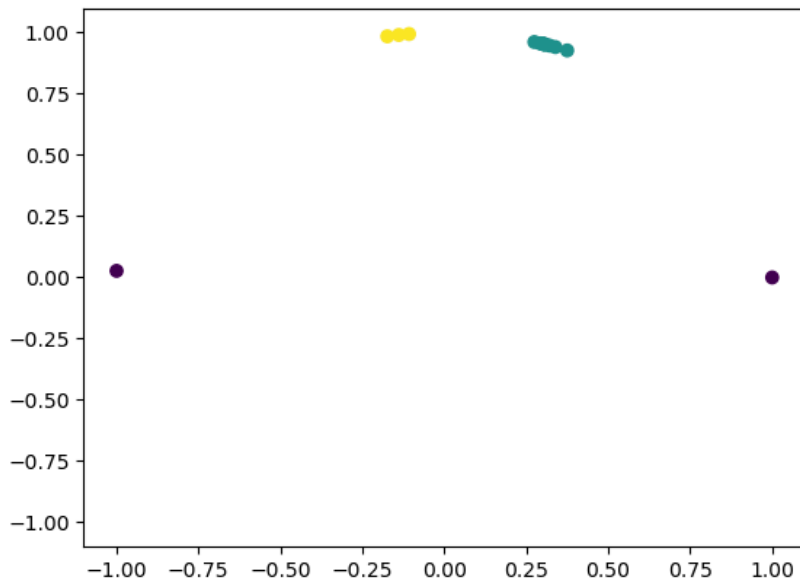
- Set of lines $L = \{\tilde{l}^{(i)}\}$

Output. Two sets of parallel lines L_1 and L_2 , where each line in L_1 is orthogonal with a line in L_2 in the original space.

Step 1. For each line $\tilde{l}^{(i)}$, compute the normalised gradient vector $\hat{v}^{(i)} = \frac{1}{\sqrt{l_1^{(i)^2} + l_2^{(i)^2}} \begin{bmatrix} l_1^{(i)} \\ l_2^{(i)} \end{bmatrix}$

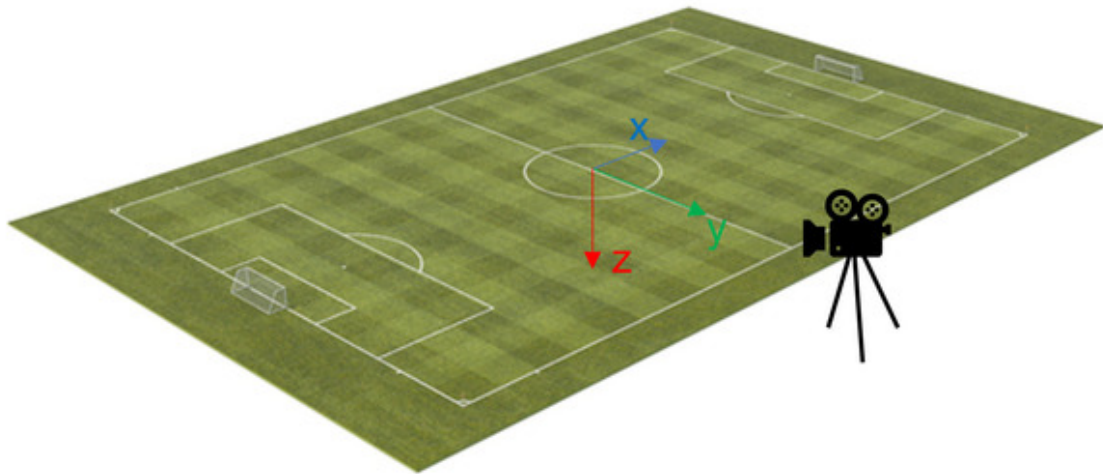
Step 2. Use a clustering algorithm $\mathcal{C}$ to group the lines into two groups.

Clustering of gradients (iv)

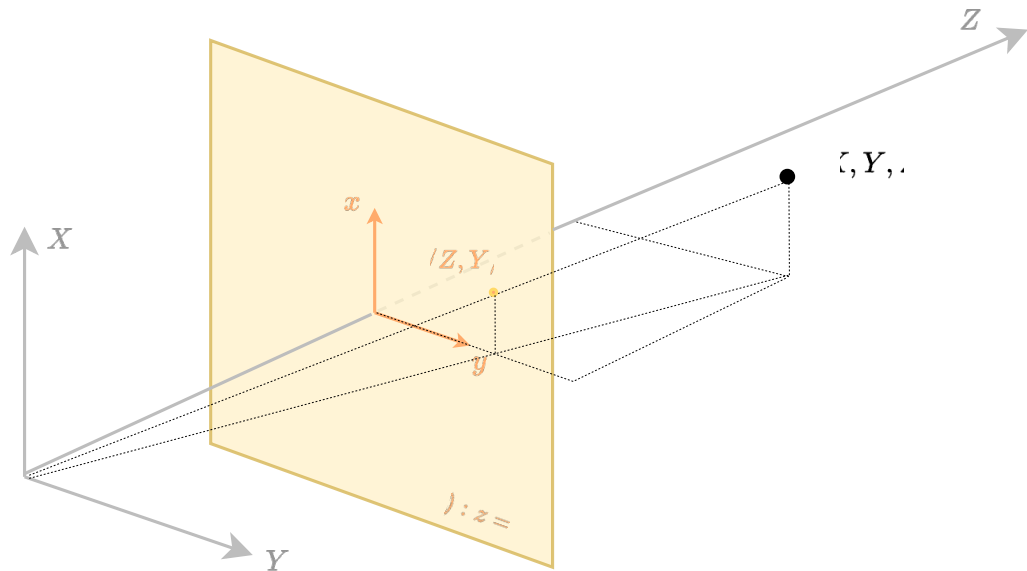


Stage 3. Homography reconstruction

SoccerNet camera model [1]



Projective geometry



Projective geometry (ii)

- Homogenous representation of a point and a line:

$$\tilde{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; \tilde{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

- Incidence:

$$(p) \in (\ell) \Leftrightarrow \tilde{l}^T \tilde{x} = 0$$

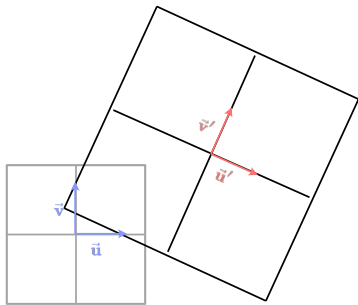
- Line passing two points:

$$\tilde{p} = \tilde{l}_1 \times \tilde{l}_2$$

- Intersection of two lines:

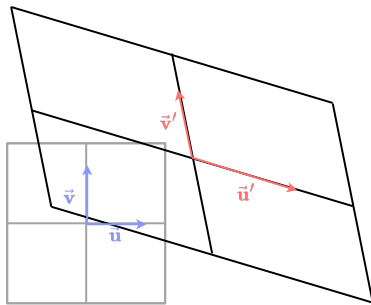
$$\tilde{l} = \tilde{x}_1 \times \tilde{x}_2$$

Transformations in 2D



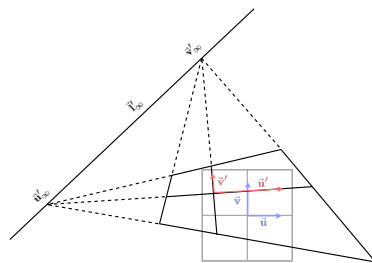
$$\tilde{\mathbf{p}}' = \begin{bmatrix} k \cos \theta & -k \sin \theta & v_x \\ k \sin \theta & k \cos \theta & v_y \\ 0 & 0 & 1 \end{bmatrix} \tilde{\mathbf{p}}$$

Similarity



$$\tilde{\mathbf{p}}' = \begin{bmatrix} a_{11} & a_{12} & v_x \\ a_{21} & a_{22} & v_y \\ 0 & 0 & 1 \end{bmatrix} \tilde{\mathbf{p}}$$

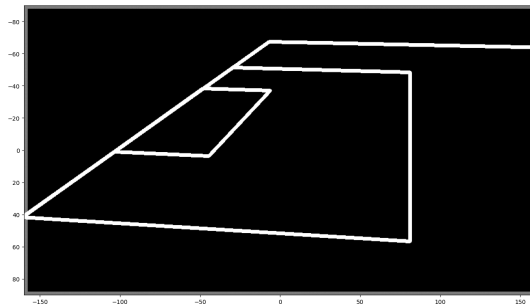
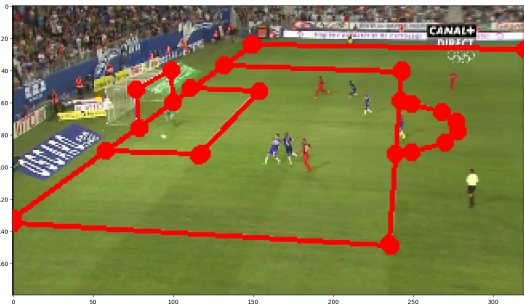
Affinity



$$\tilde{\mathbf{p}}' = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \tilde{\mathbf{p}}$$

Homography

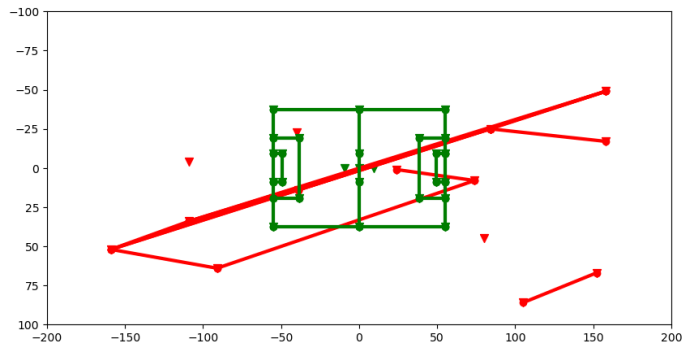
Homography reconstruction: Known correspondences



,

- Recover as many unseen keypoints as possible
- Direct Linear Transform (DLT) algorithm (*The 4-point algorithm*)

Homography reconstruction: Unknown correspondences



- Homography rectification
- Affine rectification
- Similarity rectification
- RANSAC

Homography reconstruction: Known correspondences

- Find a homography H given a set of points $P = \{\tilde{\mathbf{p}}^{(i)}\}$ and $Q = \{\tilde{\mathbf{q}}^{(i)}\}$, and that for all i :

$$H : \tilde{\mathbf{p}}^{(i)} \mapsto \tilde{\mathbf{q}}^{(i)}$$

- Need at least four point to determine such homography
- If more than four points $\rightarrow$ least squares approach $\rightarrow$ SVD

Homography reconstruction: Known correspondences (ii)

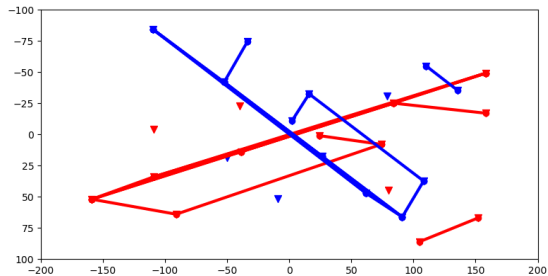
$$H\tilde{\mathbf{p}}^{(i)} \times \tilde{\mathbf{q}}^{(i)} = \tilde{\mathbf{0}}$$

$$\Leftrightarrow \begin{cases} \left(h_{11}p_1^{(i)} + h_{12}p_2^{(i)} + h_{13}p_3^{(i)} \right) q_2^{(i)} - \left(h_{21}p_1^{(i)} + h_{22}p_2^{(i)} + h_{23}p_3^{(i)} \right) q_1^{(i)} = 0 \\ \left(h_{21}p_1^{(i)} + h_{22}p_2^{(i)} + h_{23}p_3^{(i)} \right) q_3^{(i)} - \left(h_{31}p_1^{(i)} + h_{32}p_2^{(i)} + h_{33}p_3^{(i)} \right) q_2^{(i)} = 0 \\ \left(h_{31}p_1^{(i)} + h_{32}p_2^{(i)} + h_{33}p_3^{(i)} \right) q_1^{(i)} - \left(h_{11}p_1^{(i)} + h_{12}p_2^{(i)} + h_{13}p_3^{(i)} \right) q_3^{(i)} = 0 \end{cases}$$

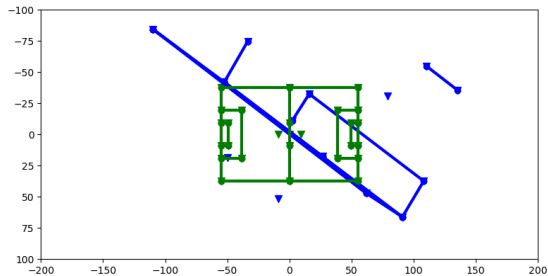
$$\Leftrightarrow \begin{bmatrix} p_1^{(i)} q_2^{(i)} & p_2^{(i)} q_2^{(i)} & p_3^{(i)} q_2^{(i)} & -p_1^{(i)} q_1^{(i)} & -p_2^{(i)} q_1^{(i)} & -p_3^{(i)} q_1^{(i)} & -p_1^{(i)} q_2^{(i)} & -p_2^{(i)} q_2^{(i)} & -p_3^{(i)} q_2^{(i)} \\ p_1^{(i)} q_3^{(i)} & p_2^{(i)} q_3^{(i)} & p_3^{(i)} q_3^{(i)} & p_1^{(i)} q_1^{(i)} & p_2^{(i)} q_1^{(i)} & p_3^{(i)} q_1^{(i)} & p_1^{(i)} q_1^{(i)} & p_2^{(i)} q_1^{(i)} & p_3^{(i)} q_1^{(i)} \\ -p_1^{(i)} q_3^{(i)} & -p_2^{(i)} q_3^{(i)} & -p_3^{(i)} q_3^{(i)} & & & & & & \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ \vdots \\ h_{31} \\ h_{32} \\ h_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Leftrightarrow A^{(i)} \tilde{\mathbf{h}} = \tilde{\mathbf{0}}$$

Homography reconstruction: Unknown correspondences



Affine and Metric rectification



Similarity rectification

Homography reconstruction: Similarity rectification

- Find a similarity H_S that maps a line segment in the source P to a line segment in the target Q .
- The mapped line segment in P might be the whole segment or a *subsegment* in Q .
- Only need **one** correspondence of segments from P to $Q \Rightarrow$ exhaustive search.
- Which correspondence is the best performing one? How should a correspondence be evaluated?

Homography reconstruction: Similarity rectification

Similarity rectification from a correspondence of two segments.

- *Input:* Four points $\vec{p}^{(1)}, \vec{p}^{(2)}, \vec{q}^{(1)}, \vec{q}^{(2)}$ indicating two segments $\vec{p}^{(1)}, \vec{p}^{(2)}$ and $\vec{q}^{(1)}, \vec{q}^{(2)}$.
- *Output:* The similarity H_S .

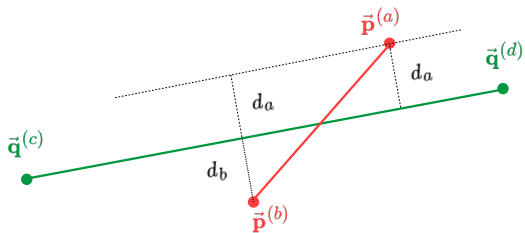
Step 1. Let $(p_1^{(i)}, p_2^{(i)}, 1) := \tilde{p}^{(i)}, (q_1^{(i)}, q_2^{(i)}, 1) := \tilde{q}^{(i)}$ for $i = 1, 2$.

Step 2. Solve the linear system $H_S \tilde{p}^{(i)} = \tilde{q}^{(i)}$.

$$\begin{cases} ap_1^{(1)} - bp_2^{(1)} + c = q_1^{(1)} \\ ap_2^{(1)} + bp_1^{(1)} + d = q_2^{(1)} \\ ap_1^{(2)} - bp_2^{(2)} + c = q_1^{(2)} \\ ap_2^{(2)} + bp_1^{(2)} + d = q_2^{(2)} \end{cases} \quad \therefore \quad \begin{bmatrix} p_1^{(1)} & -p_2^{(1)} & 1 & 0 \\ p_2^{(1)} & p_1^{(1)} & 0 & 1 \\ p_1^{(2)} & -p_2^{(2)} & 1 & 0 \\ p_2^{(2)} & p_1^{(2)} & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} q_1^{(1)} \\ q_2^{(1)} \\ q_1^{(2)} \\ q_2^{(2)} \end{bmatrix}$$

The similarity can thus be solved using any linear solver such as the SVD.

Homography reconstruction: Similarity rectification



Homography reconstruction: Similarity rectification

Modified RANSAC to find segment correspondence.

Step 1. Let $S_P := \emptyset$ and $S_Q := \emptyset$

Step 2. Select a random segment $\mathbf{p}^{(\alpha\beta)} = (\bar{\mathbf{p}}^{(\alpha)}, \bar{\mathbf{p}}^{(\beta)})$ in $\mathbf{P}$ and a random segment $\mathbf{q}^{(\gamma\delta)} = (\bar{\mathbf{q}}^{(\gamma)}, \bar{\mathbf{q}}^{(\delta)})$.

Step 3. Find a similarity H_S such that the segment $\mathbf{p}^{(\alpha\beta)}$ is mapped to $\mathbf{q}^{(\gamma\delta)}$.

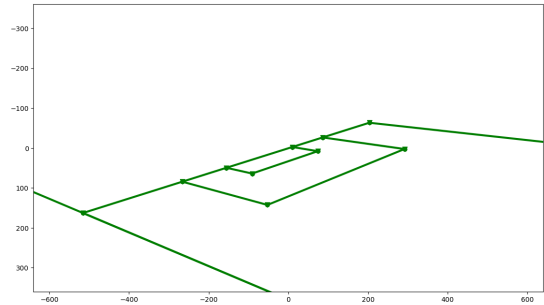
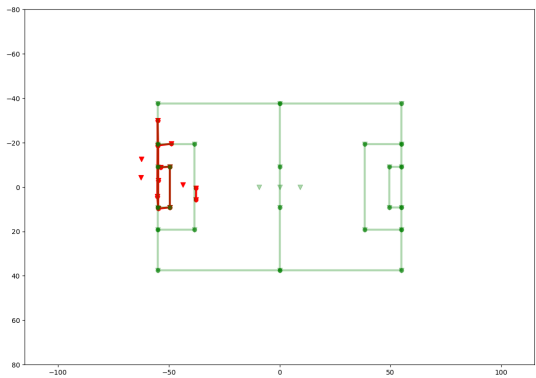
Step 4. For each $\mathbf{p}^{(ab)}$ in $\mathbf{P}$ and $\mathbf{q}^{(cd)}$, calculate the distances between the endpoints and the other segment:

$$d_a := \text{dist}(\bar{\mathbf{p}}^{(a)}, \mathbf{q}^{(cd)}); \quad d_b := \text{dist}(\bar{\mathbf{p}}^{(b)}, \mathbf{q}^{(cd)})$$

• Let $d_{\text{avg}} = \frac{1}{2}(d_a + d_b)$. If $d_{\text{avg}} \leq T$, $S_P := S_P \cup \{ab\}$, $S_Q := S_Q \cup \{cd\}$.

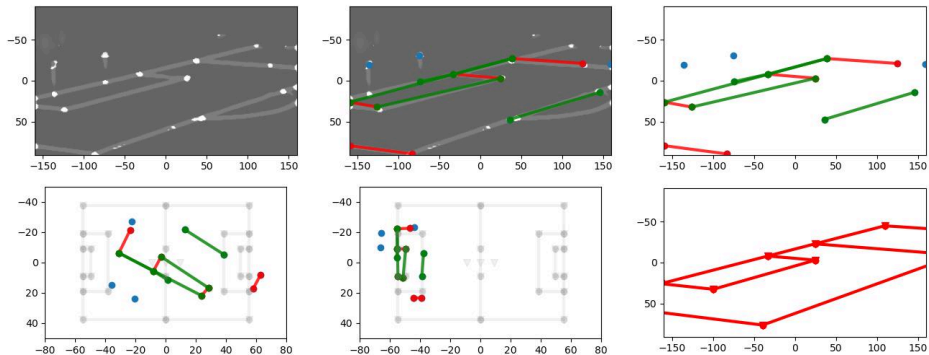
Step 5. Let $\text{Score} = |S_P| + |S_Q|$. If $\text{Score} \geq \text{Score}_{\text{Accept}}$, return H_S . Otherwise, repeat the process.

Results



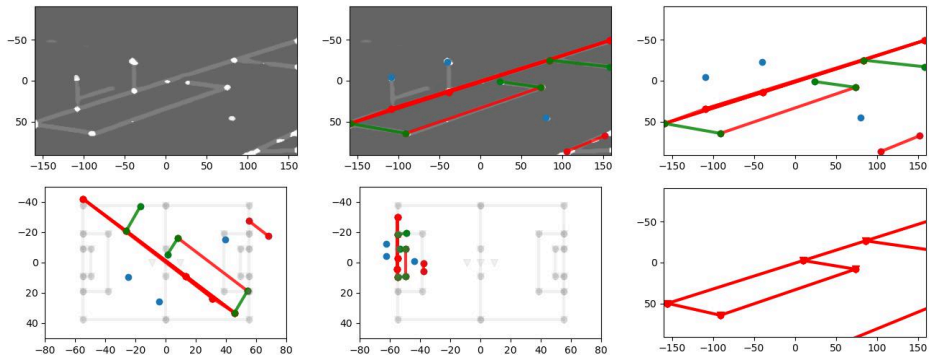
Results (ii)

00126.jpg



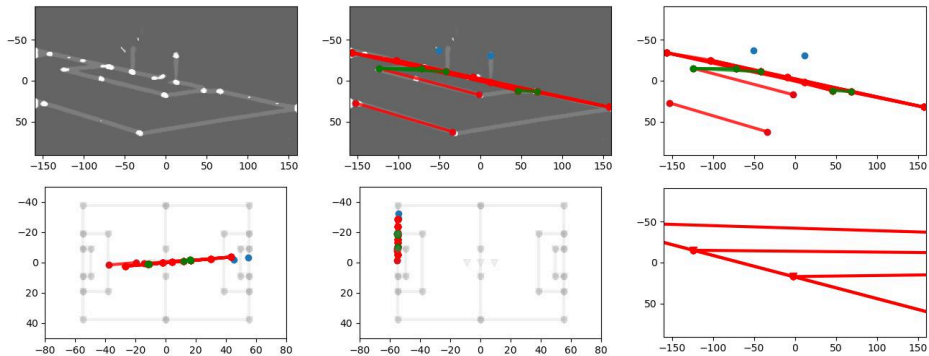
Results (iii)

01210.jpg



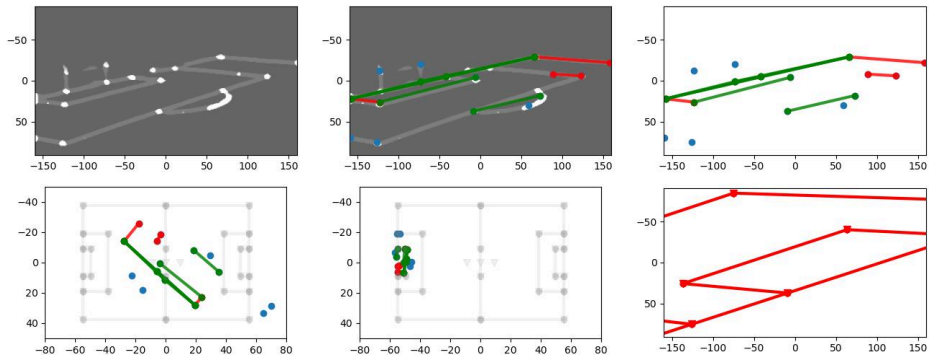
Results (iv)

00132.jpg



Results (v)

02094.jpg



References

- [1] “SoccerNet - 2025.” Accessed: July 26, 2025. [Online]. Available: <https://www.soccer-net.org/challenges/2025>
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